Integrated Performance of FRP Tendons with Fiber Hybridization

Nageh M. Ali[1]; Xin Wang[2]; and Zhishen Wu, M.ASCE[3]

Abstract: A fundamental understanding of the mechanical properties and the failure mechanism of hybrid fiber-reinforced polymers (FRP) is required for the effective application of FRP in construction. This paper presents a new methodology for predicting the tensile behavior of hybrid FRP tendons by considering the interfacial stress transfer between the resin and the fibers in hybrid FRP. Subsequently, the authors utilize the fundamental concepts of fracture mechanics to derive a model capable of predicting the mechanical properties of hybrid FRPs. For this paper, the authors conducted an experimental study on the tensile properties of hybrid basalt/carbon FRP tendons and hybrid glass/carbon FRP tendons. They identified the effects of resin type, fiber fraction, and fiber arrangement over the cross section. The results show that the stress-strain relationship of hybrid FRP can be modified from the linear behavior of FRP to a ductile behavior with a steady pseudoyielding plateau and a high ultimate failure strain. Meanwhile, the load drop in hybrid FRP, which is attributable to the fracture of the fibers with low elongation capacity, can be controlled effectively by the proper design of the hybrid fiber proportions, resin type, and volume fraction. The proposed hybridization results in improving the deformation ability of fibers with low elongation capacity. Moreover, the proposed model for the description of failure progression and prediction of mechanical properties is verified by good agreement with the experimental results of this paper and those from prior studies by others. **DOI: 10.1061/(ASCE)CC.1943-5614.0000427.** *© 2013 American Society of Civil Engineers.*

Author keywords: Carbon fiber; Basalt fiber; Hybrid tendons; Iterative model; Interfacial stress.

Introduction

Fiber-reinforced polymer (FRP) composites have become candidate materials for civil engineering applications because of their superior advantages such as a high strength-to-weight ratio, corrosion resistance, a wide range of working temperatures, and the ease of handling. FRPs have been used externally to improve the flexural and shear capacities of beams, slabs, and shear walls effectively (Coronado and Lopez 2006; Eid and Paultre 2008; Sobhy and Khalid 2006; Wu et al. 2007). Moreover, they can be used as reinforcement tendons, replacing conventional steel reinforcement in reinforced concrete (RC) or long-span structures (Aiello and Ombers 2002; Benmokrane et al. 1996; Yan et al. 2010; Wang and Wu 2010). However, the application of FRPs in structures cannot yet satisfy all of the structural integrity requirements. For instance, in the case of FRP with high strength and high elastic modulus, such as carbon FRP, the desired ductility of the structure is difficult to achieve. This is attributable to the inherently small failure strain of carbon FRP. Meanwhile, carbon FRP is also more expensive than other types of FRP. In contrast, the relatively inexpensive FRPs, such as glass FRP with larger failure strain, can make structures more ductile; however, their relatively low modulus usually restricts their application because of structural deformation requirements. To overcome the aforementioned limitations and to enhance the utilization of various FRP composites, this paper introduces the hybridization of different types of fibers to overcome their shortcomings, to integrate their advantages, and to subsequently achieve the best performance-to-price ratio.

Previous studies have mainly focused on the hybridization of carbon/PBO (polypara-phenylene-benzo-bis-oxazole) fibers with glass fibers (Bakis et al. 2001; Wu 2004; Kentaro et al. 2007). Such hybrid FRPs were initially introduced in the 1970s by focusing on the integration of carbon and glass FRPs, which resulted in residual compressive stress in the carbon fibers to improve the overall structural performance of the entire FRP system. By focusing on developing applications for structural strengthening, Bakis et al. (2001) and Wu (2004) further developed the mechanism of hybrid FRPs for both FRP tendons and sheets. These studies revealed that in both FRP tendons and sheets, the high elongation (HE) fibers in hybrid FRP could undertake the load released by the fractured low elongation fibers so that damage localization and catastrophic failure could be avoided in the hybrid FRP composites. Kentaro et al. (2007) further suggested a control index of stress drop for designing the pseudoductility of hybrid FRP sheets with PBO fiber and glass fiber. Aside from the previously mentioned studies, Young et al. (2007) and Tamuzs et al. (1996) investigated the hybrid effect from the perspective of fiber distribution, resin types, and structural composition. To the best of the authors' knowledge, the hybrid effect of carbon and basalt fiber rods have not been addressed in previous studies. Basalt fiber composites are newly-developed materials that not only display higher strength and elastic modulus but also have a similar cost and better chemical stability compared with E-Glass FRP composites (Wang and Wu 2010). The material and structural behavior of basalt FRPs also show enhancement by hybridizing with carbon fibers, such as the improvement of fatigue behavior, performance under elevated

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temperature, and behavior under freeze-thaw actions. Thus, basalt fibers can be regarded as a potential alternative material, if hybridized with carbon fibers, to achieve superior behavior compared with glass fibers. This paper addresses the hybridization of carbon and basalt fibers.

The failure mechanism and prediction of the mechanical properties of hybrid FRP composites have not been sufficiently addressed in previous studies. To the best of the authors' knowledge, the prediction of the tensile performance of hybrid composites that has been discussed in previous studies has mainly focused on the simple rule-of-mixtures. Considering two-fiber hybrid composites, the mixture law assumes that the hybrid components work together as an integrated single media up to failure of the fiber with the lowest elongation capacity, and the modulus is defined by multiplying the modulus of each component by its volume fraction. This approach relies on the assumption that the failure of each fiber is reflected in calculating the stiffness and strength. However, recent research has revealed that although this method may predict the modulus of hybrid FRP accurately, it is not able to define the load loss at rupture of the lowest elongation fiber nor can it predict the load and strain at final rupture.

To address the aforementioned shortcomings, this paper will first discuss a new analytical model for predicting the tensile behavior of a hybrid FRP, which takes into account the interfacial stress transfer between the hybridization components after low elongation (LE) fiber rupture. This paper investigates the mechanical behavior of basalt/carbon and glass/carbon hybrid FRP tendons and discusses the influence of resin type, fiber fraction, and fiber arrangement. Finally, the authors will verify the prediction model through a comparison with experimental data and other models.

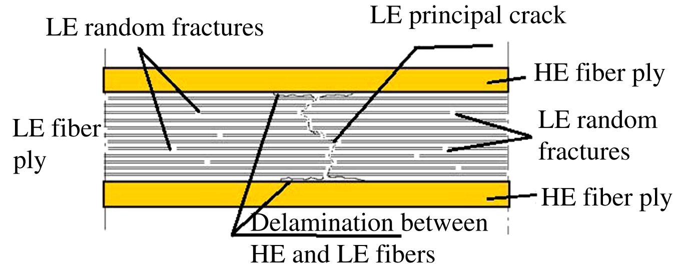


Fig. 1. Random failure of LE fibers and delamination cracks between LE fibers and HE fibers

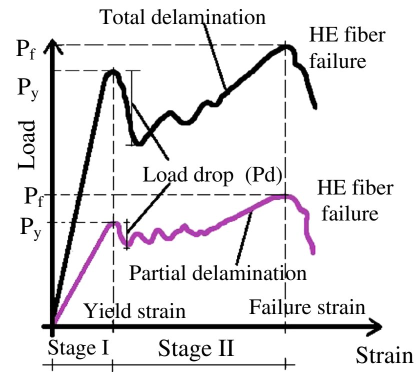


Fig. 2. Proposed tensile behavior of hybrid FRP composite

Proposed Model for Predicting Tensile Performance of Hybrid FRP Composites

Failure Mechanism

To predict the mechanical behavior of hybrid FRP tendons, the authors first investigated the mechanics of fracture and its progression. Manders and Bader (1981) reported that the load-strain curve of an alternating glass/carbon hybrid composite was virtually linear up to the first failure of carbon fibers. With increasing strain, the first macroscopic fracture occurred at the central carbon ply; the principal transverse crack ran across the entire cross section of the carbon ply. A delamination crack generated at the intersection between the inner carbon ply and the outer glass plies. Fig. 1 shows a schematic drawing of a glass/carbon hybrid FRP specimen subjected to a strain that caused carbon fiber rupture (Michael and Bader 1987). The figure indicates the random failure of LE fiber and the longitudinal delamination cracks between the glass and carbon fibers.

Based on the aforementioned observations in prior studies, the authors proposed load-strain curves of hybrid FRP tendons/sheets as shown in Fig. 2 for this paper. For each case, the curves were divided into two stages. In the first stage, up to the failure strain of the LE fibers, the hybrid composite worked as a single unit, the curve was virtually linear, and the modulus could be accurately determined by the simple rule-of-mixtures. In the second stage, with increasing strain, LE filaments broke randomly, causing transverse cracking in the matrix and debonding between the LE filaments and the surrounding resin. Meanwhile, delamination cracks at the interface between LE fibers and HE fibers also occurred, as shown in Fig. 1. The propagation of delamination between the LE and HE fibers increased with an increasing volume ratio of LE fibers. Total delamination occurred when the load at LE fiber rupture was very high. Two types of load-strain behavior for hybrid composites were possible, as shown in Fig. 2, depending on whether there was partial delamination or total delamination. For the partial delamination case, the authors assumed that the stress in the HE fibers at the location of the delamination was higher than the stress in the HE fibers far away from the crack location. This phenomenon occurred, because the bond between the LE and HE fibers was still present, and both materials worked together to carry the load away from the crack location. In contrast, the bond at the crack was broken, and thus, only the HE fibers carried the load. The magnitude and the extent of delamination increased when the load increased, and the contribution of the LE fibers to the stiffness of the hybrid FRP decreased nonlinearly. The presence of several load drops in the load-stain curve was attributable to the random failure of the LE fibers and the sudden increase in the delamination length. The sudden drop in load could cause partial damage of the HE fibers, reducing the total failure load capacity. Finally, when total debonding between the LE and HE fibers occurred, only the HE fibers resisted the load so that the load-strain curve remained linear up to total rupture of the composite. For the total delamination case, the LE fibers did not contribute to the hybrid stiffness and the drop in load was larger than that of the partial delamination case. As the load was carried only by the HE fibers, the load-strain curve was linear up to failure. The rule-of-mixtures method for predicting the hybrid behavior could only be used in the case of total delamination.

Therefore, the authors proposed a model that predicts the tensile behavior of hybrid FRP composites on the basis of considering the contribution of the LE fibers to the hybrid composite after its failure and that can be achieved by computing the stress distribution over the entire length of the HE fibers after rupture of the HE fibers on the basis of the principles of fracture mechanics.

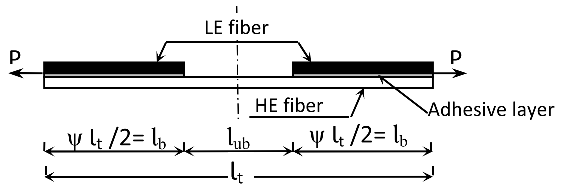


Fig. 3. Details of two-fiber hybrid tensile test specimen after LE fiber rupture

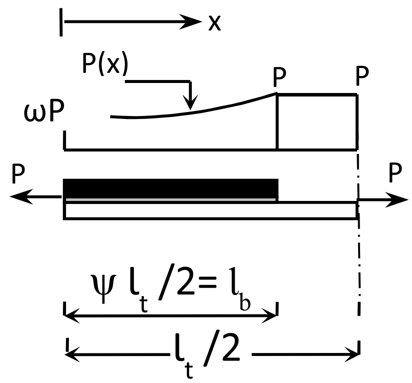


Fig. 4. Load distribution carried by HE fiber after failure of the LE fiber

Simulation of Interfacial Stress Transfer in Hybrid FRP Composites

To find the contribution of the LE fiber to the hybrid stiffness after its total rupture, the authors studied the interfacial stress/load transfer from the HE fibers to the LE fibers. Fig. 3 shows the details of a tensile test specimen for a hybrid composite consisting of LE and HE fibers after LE fiber rupture. The specimen length (l_t) was divided into two parts: the bonded part (l_b), where LE fibers were still bonded to the HE fibers, and the unbonded part (l_{ub}), where the LE and HE fibers were totally delaminated. In the bonded part, some load transferred through the adhesive layer (resin matrix) from the HE fibers to the LE fibers, and then the load carried by the LE fibers decreased from the left-end of the left-half of the specimen, as shown in Fig. 4. With increasing load, debonding at the interface propagated from the right-end of the left-half of the specimen shown in Fig. 4, and the bonded length ratio ($\psi = l_b/l_t$) decreased until total debonding was achieved. The authors assumed that the shear stress-slip relationship was linear before the occurrence of interfacial fracture and the value of shear stress suddenly dropped to zero when the slip exceeded its maximum value without consideration of softening. Therefore, the specimen behaved linearly until total debonding. Consequently, by using the principle of superposition, the load distribution between the LE and HE fibers could be simply divided into two common adhesive joints, namely the pull-pull joint and the pull-push joint as illustrated in Fig. 5, where ω is the proportion of load that would transfer to LE fibers at the left-end of the specimen.

Fundamental Interface Models

Considerable research has been carried out regarding how to solve the interfacial-shear stress transfer in FRP-strengthened structures, especially for FRP-to-concrete adhesively bonded joints. Yuan et al. (2001), Wu et al. (2002), and Yuan et al. (2004) introduced several

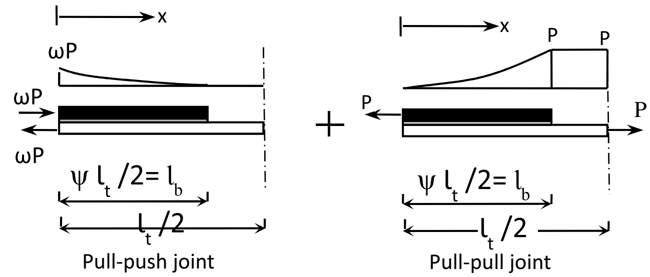


Fig. 5. Load distribution carried by HE fiber after failure of LE fiber divided by superposition into two common joints

nonlinear interfacial constitutive laws describing the precracking and postcracking behavior of the FRP-concrete interface to solve the nonlinear interfacial short-term stress transfer problem. In this paper, the authors adopted and modified a nonlinear constitutive law to solve for the interfacial-shear stress transfer in a two-fiber hybrid sheet specimen subjected to tension.

As previously mentioned, the authors simplified the analysis of a two-fiber hybrid sheet subjected to tension into two common adhesive joints, as shown in Fig. 5, to understand the stress transfer and debonding behavior. The adhesive layer was subjected to shear only. The area and the modulus of the HE and LE fibers were represented as A_1, A_2, E_1, and E_2, respectively. The width of the interfacial surface was represented as S, and the bonded length of the specimen-half was given by ($\psi \cdot l_t/2$). For a bonded joint with two adherent materials, that is, the LE and HE fibers in the context of this paper, subjected to axial deformation only and with the bonded interface subjected to pure shear only, the following governing equations may be derived (Wu et al. 2002):

$$\frac{d^2\delta}{dx^2} - \frac{2G_f}{\tau_f^2}\lambda^2 f(\delta) = 0 \tag{1}$$

where for the pull-pull joint, the normal stress in the HE fibers is

$$\sigma_1 = \frac{\tau_f^2 \bullet S}{2G_f A_1 \lambda^2}\left(\frac{d\delta}{dx} + \frac{P}{E_2 A_2}\right) \tag{2}$$

and for the pull-push joint

$$\sigma_1 = \frac{\tau_f^2 \bullet S}{2G_f A_1 \lambda^2}\frac{d\delta}{dx} \tag{3}$$

where

$$\lambda^2 = \frac{\tau_f^2 \bullet S}{2G_f}\left(\frac{1}{E_1 A_1} + \frac{1}{E_2 A_2}\right) \tag{4}$$

where δ = the relative displacement (or slip) between adherents, τ_f = the local bond strength, and G_f = the interfacial fracture energy.

Theoretical Analysis

Using appropriate boundary conditions and following procedures similar to those found in Yuan et al. (2001) and Wu et al. (2002), the relative displacement δ, adhesive layer shear stress τ, and normal stress of the HE fiber σ_1, are given by the following equations for the pull-pull joint:

$$\delta = \left[\frac{P}{\lambda E_1 A_1 \sinh(\lambda\psi l_t/2)} + \frac{P}{\lambda E_2 A_2 \tanh(\lambda\psi l_t/2)}\right] \bullet \cosh(\lambda x) - \frac{P}{\lambda E_2 A_2}\sinh(\lambda x) \quad (5)$$

$$\tau = \left\{\frac{P}{\lambda}\left[\left(\frac{1}{E_1 A_1 \sinh(\lambda\psi l_t/2)} + \frac{1}{E_2 A_2 \tanh(\lambda\psi l_t/2)}\right) \bullet \cosh(\lambda x) - \frac{1}{E_2 A_2}\sinh(\lambda x)\right]\right\} \bullet \frac{\tau_f^2}{2G_f} \quad (6)$$

$$\sigma_1 = \left\{\left[\left(\frac{1}{E_1 A_1 \sinh(\lambda\psi l_t/2)} + \frac{1}{E_2 A_2 \tanh(\lambda\psi l_t/2)}\right) \bullet \sinh(\lambda x) - \frac{1}{E_2 A_2}\cosh(\lambda x) + \frac{1}{E_2 A_2}\right]\right\} \frac{\tau_f^2 \bullet S \bullet P}{2G_f A_1 \lambda^2} \quad (7)$$

and the same are given by the following equations for the pull-push joint:

$$\delta = \frac{-\omega P}{S} \bullet \frac{2G_f \lambda}{\tau_f^2}\left[\frac{\cosh(\lambda x)}{\tanh(\lambda\psi l_t/2)} - \sinh(\lambda x)\right] \quad (8)$$

$$\tau = \frac{\omega P \lambda}{S}\left[\frac{-\cosh(\lambda x)}{\tanh(\lambda\psi l_t/2)} + \sinh(\lambda x)\right] \quad (9)$$

$$\sigma_1 = \frac{\omega P}{A_1}\left[\frac{-\sinh(\lambda x)}{\tanh(\lambda\psi l_t/2)} + \cosh(\lambda x)\right] \quad (10)$$

For the boundary condition at $x = 0$

$$(\delta)_{\text{total}} = (\delta)_{\text{pull-pull}} + (\delta)_{\text{pull-push}} = 0 \quad (11)$$

ω is determined to be

$$\omega = \frac{\tau_f^2 \bullet S}{2G_f \lambda^2 E_1 A_1}\left[\frac{1}{\cosh(\lambda\psi l_t/2)} + \frac{1}{\beta}\right] \quad (12)$$

where

$$\beta = \frac{E_2 A_2}{E_1 A_1} \quad (13)$$

Substituting ω from Eq. (12) into Eqs (8)–(10) and using the principle of superposition, the relative shear displacement δ, the adhesive layer shear stress τ, and the normal stress σ_1 of the HE fibers may be obtained from the following equations:

$$\delta = \frac{P}{\lambda E_1 A_1 \cosh(\lambda\psi l_t/2)} \bullet \sinh(\lambda x) \quad (14)$$

$$\tau = \frac{\tau_f^2}{2G_f}\left[\frac{P}{\lambda E_1 A_1 \cosh(\lambda\psi l_t/2)} \bullet \sinh(\lambda x)\right] \quad (15)$$

$$\sigma_1 = \frac{\tau_f^2 PS}{2G_f \lambda^2 E_1 A_1}\left[\frac{\cosh(\lambda x)}{A_1 \cosh(\lambda\psi l_t/2)} + \frac{1}{A_1 \beta}\right] \quad (16)$$

Under small loads, there is no interfacial debonding as long as the interfacial-shear stress at the right-end of the specimen is less than τ_f. The maximum transferable force occurs when the shear stress reaches the value τ_f at the right-end. Once debonding at the right-end develops, the peak shear stress τ_f moves towards the left-end and the bonded length ratio (ψ) reduces. Therefore, when the interfacial-shear stress at the right-end ($x = \psi \cdot l_t/2$) reaches its maximum value τ_f, the maximum transferable load is obtained by

$$P_{\text{dep}} = \frac{2G_f E_1 A_1 \lambda}{\tau_f \tanh(\lambda\psi l_t/2)} \quad (17)$$

which is accompanied by the propagation of debonding. From Eq. (17), one can conclude that for further reduction in the bonded length ratio (ψ), increasing the applied load is required. Moreover, when the bonded length ratio goes to zero, the applied load required for further debonding goes to infinity. As a result, the strength of the HE fibers governs total failure, and the failure load (P_f) can be computed by the following:

$$P_f = \phi \bullet E_1 \varepsilon_1 A_1 \quad (18)$$

where ϕ = a reduction factor as a result of the load drops at LE fiber rupture, which can be taken as 0.9. Furthermore, total debonding is not achieved and the minimum bonded length ratio ($\psi_{\min}$) corresponding to the failure load can be computed by

$$\psi_{\min} = \frac{2}{\lambda l_t} \arctan h\left(\frac{\delta_f \lambda}{0.9 \bullet \varepsilon_1}\right) \quad (19)$$

where ε_1 is the failure strain of the HE fibers.

Finally, as the distribution of normal stress carried by the HE fibers is uniform only at the unbonded part, the total strain corresponding to the applied load can be evaluated by integrating the normal stress of the HE fiber. Therefore, Eq. (20) completes the relationship between the applied load, bonded length ratio, and total strain corresponding to the applied load as shown

$$\varepsilon = \frac{P(1-\psi)}{E_1 A_1} + \frac{\tau_f^2 P \bullet S}{G_f \lambda^2 E_1^2 A_1 l_t}\left[\frac{\tanh(\lambda\psi l_t/2)}{A_1 \lambda} + \frac{\psi l_t}{2A_1 \beta}\right] \quad (20)$$

To summarize the theoretical derivation, the load-strain relationship of a hybrid tendon-sheet can be determined by using the following steps according to the relevant state of the LE fibers:

1. Before LE fiber rupture, following the rule-of-mixtures, the modulus of elasticity (E) and pseudoyielding load (P_y) can be predicted from the following equations:

$$E = E_1 v_1 + E_2 v_2 \quad (21)$$

$$P_y = f_1 A_1 + \varepsilon_1 E_2 A_2 \quad (22)$$

   where v_1 and v_2 = the volume ratio of HE and LE fibers, respectively, and f_1 = the tensile strength of the LE fibers.
2. After LE fiber rupture, the following steps are taken:
   a. Find the minimum ratio of the bonded length ($\psi_{\min}$) by using Eq. (19);
   b. Select a bonded length ratio (ψ_i) in the range of $\psi_{\min} \le \psi_i \le 1.0$;
   c. From Eq. (17), find the maximum load required to further debonding for the selected ψ_i;
   d. Utilize Eq. (20) to find the strain corresponding to the applied load;
   e. Repeat steps *b.* through *d.* for different values of ψ_i; and
   f. Draw the relationship between load and strain.

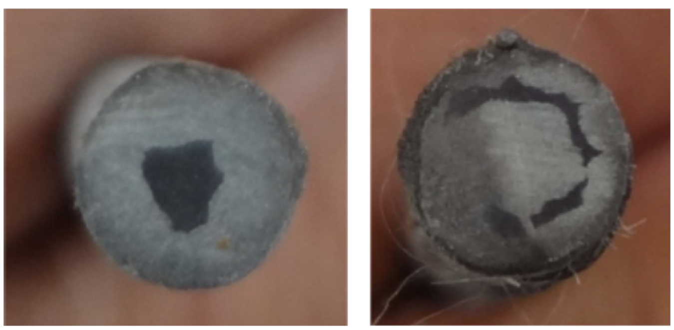
Experimental Studies

Material and Specimen Specifications

For this paper, the authors produced basalt, carbon, glass, hybrid basalt/carbon (B/C), and hybrid glass/carbon (G/C) FRP tendons with 6-mm diameter by using a pultrusion process. They designed B/C FRP tendons with five different basalt and carbon volume proportions and used three different glass and carbon volume proportions for the hybrid G/C FRP tendons. In the hybrid B/C FRP tendon specimens, they considered two types of resins (epoxy and vinyl-ester), three fiber volume fractions V_f (50, 60, and 70%), and two types of fiber distribution (centralized and dispersed) (Fig. 6). Table 1 summarizes the specimen details. Table 2 shows the mechanical properties of the fibers studied in this paper, according to the manufacturer's data.

Sample Preparation and Test Setup

The authors prepared a total of 105 samples. They tested five specimens of each FRP tendon. They cut the specimens to a length of 840 mm, with an anchorage length of 300 mm at each end. They used sand blasting at the anchor ends to improve the bond between the tendons and the anchorage device. They used steel tubes with 10-mm inner diameter and 2-mm wall thickness as anchorage devices. They used epoxy resin to bond the tubes to the tendons. They tested the specimens 10 days after installing the anchorage. All tests were conducted on a universal testing machine, maintaining a loading rate of 3 mm/ min. They measured the strain by means of an LVDT extensometer with a 100-mm gauge length placed at the midlength of the test coupon.



(a) (b)

Fig. 6. Carbon arrangement in hybrid tendons

Table 1. Details of FRP Tendon Specimens

Group number	Specimen ID	Fiber volume ratio (V_f%)	Basalt or glass / Carbon ratio ($\{V_B$ or $V_G\}/V_C$)	Resin type	Carbon arrangement
1	CFRP-E	56.6	—	Epoxy	All
	CFRP-V	56.6	—	Vinylester	All
	BFRP-E	57.7	—	Epoxy	—
	BFRP-V	57.7	—	Vinylester	—
	GFRP-E	58.6	—	Epoxy	—
	GFRP-V	58.6	—	Vinylester	—
2	B1C1-E	57.7	1.0	Epoxy	Centered
	B1C1-V	57.7	1.0	Vinylester	Centered
	B3C1-E	57.9	2.7	Epoxy	Centered
	B3C1-V	57.9	2.7	Vinylester	Centered
	B4C1-E	58.0	3.6	Epoxy	Centered
	B4C1-V	58.0	3.6	Vinylester	Centered
	B6C1-V	59	6.5	Vinylester	Centered
	B8C1-V	57.5	8.2	Vinylester	Centered
3	B4C1-V-D	58	3.6	Vinylester	Dispersed
	B6C1-V-D	59	6.5	Vinylester	Dispersed
4	B6C1-V-50%	48	6.6	Vinylester	Centered
	B6C1-V-70%	70.3	6.5	Vinylester	Centered
5	G3C1-V	55.5	2.5	Vinylester	Centered
	G4C1-V	59	3.7	Vinylester	Centered
	G6C1-V	60.9	6.8	Vinylester	Centered

Table 2. Mechanical Properties of FRP Fibers Reported by Manufacturers

Materials	Tensile strength (MPa)	Tensile modulus (GPa)	Rupture strain (%)
Carbon	3,400	230	1.48
Basalt	2,100	91	2.31
Glass	1,500	73	2.05
Vinylester resin	79	3.7	—
Epoxy resin	51.9	3.43	—

Experimental Results

Nonhybrid Tendons

Table 3 summarizes the mechanical properties of the nonhybrid specimens including the BFRP, CFRP, and GFRP tendons. The stress-strain relationship of the specimens exhibited a perfectly linear behavior with the occurrence of brittle fracture. All specimens failed within the gauge length.

In general, tendons produced with epoxy resin (E-tendons) showed a higher performance compared with those with vinyl-ester resin (V-tendons), particularly for the case of carbon tendons. As shown in Table 3, the tensile strength, elastic modulus, and ultimate strain of the carbon E-tendons were 37.8, 5.1, and 27.6% higher than those of the carbon V-tendons. For the basalt E-tendons, the tensile strength, modulus of elasticity, and failure strain increased by 10.9, 2.1, and 1.9% in comparison with those for the basalt V-tendons. Similar to carbon E-tendons and basalt E-tendons, the glass E-tendons indicated tensile strength and failure strain that were 3.4 and 9.2% higher than those of the glass V-tendons. Moreover, for the same type of FRP, the coefficient of variation (COV) of tensile strength was relatively lower for the specimens using the epoxy resin matrix. The difference between the behavior of the vinyl-ester tendon and the epoxy tendon is likely because of the differences in characteristics and performance of the resin to redistribute loads between the fibers (Young et al. 2007). These observations indicate that epoxy resin is more compatible with both basalt and carbon fibers.

Table 3. Test Results for Nonhybrid Tendon Specimens

	Tensile strength		Elastic modulus		Ultimate strain
Specimen ID	(MPa)	Coefficient of variation (%)	(GPa)	Coefficient of variation (%)	(%)
CFRP-E	2,260	3.3	144	0.9	1.57
CFRP-V	1,640	4.3	137	0.6	1.23
BFRP-E	1,530	1.8	49	0.2	3.20
BFRP-V	1,380	3.4	48	0.3	3.14
GFRP-E	1,210	3.0	47	1.13	2.62
GFRP-V	1,170	6.6	48	4.4	2.40

Table 4. Test Results of Hybrid Tendons

Specimen ID	P_y (kN)	σ_y (MPa)	ε_y (%)	E (GPa)	P_d (kN)	P_f (kN)	σ_f (MPa)	ε_f (%)
B1C1-E	46	1,630	1.71	96	34	12	424	2.61
B1C1-V	39	1,380	1.45	94	29	10	354	1.96
B3C1-E	36	1,270	1.76	73	15	26	920	2.66
B3C1-V	29	1,030	1.50	68	10	25	884	2.33
B4C1-E	33	1,170	1.80	65	9	31	1,100	2.74
B4C1-V	26	920	1.48	64	6	25	884	2.42
B6C1-V	28	990	1.69	60	4.2	35	1,240	3.01
B8C1-V	26	920	1.76	55	3.6	35	1,240	3.61
B4C1-V-D	31	1,100	1.65	66	13.2	39	1,380	2.66
B6C1-V-D	37	1,310	2.27	60	8.4	31	1,100	2.52
B6C1-V-50%	—	—	—	47	—	19	972	1.45
B6C1-V-70%	30	1,060	1.65	70	2.9	36	1,270	2.59
G3C1-V	—	—	—	61	—	19	972	1.28
G4C1-V	20	707	1.24	57	5.1	22	778	2.34
G6C1-V	20	707	1.31	55	3.1	24	849	2.55

Note: E = modulus of elasticity of the tendon; E_f = failure strain; P_d = load drop at carbon rupture; P_f/σ_f = failure load/stress; P_y/σ_y = pseudo-yielding load/stress; and ε_y = pseudo-yielding strain.

Hybrid Tendons

Table 4 summarizes the results of the tensile tests on the hybrid tendon specimens. The pseudoyielding load (P_y) is the load corresponding to the initial rupture of the carbon fiber when the load-strain curve shifts from linearity. The load drop (P_d) is defined by the maximum loss of load during carbon failure, and the failure load (P_f) is the maximum load resisted by the tendon following the pseudoyielding stage. The load-strain behavior of the B1C1-V tendons was linear up to a strain of 1.45%, which was followed by a sudden load drop of 29 kN. The load then increased slightly before the final rupture. Load-strain curves of the hybrid E-tendons, B1C1-E, B3C1-E, and B4C1-E, had the same trend as that of the hybrid V-tendons. Specimens B6C1-V-50% and G3C1-V exhibited brittle load-strain curves with an identical failure load of 19 kN and failure strains of 1.45 and 1.28%, respectively. Figs. 7–15 show the tensile behavior of the remaining hybrid FRP tendons. The authors observed a distinct pseudoductile behavior for specimens with higher basalt ratio. The following section discusses the effect of resin type, B/C ratio, fiber volume ratio, and fiber arrangement on the mechanical properties of B/C hybrid tendons.

American Concrete Institute Committee 400 (2004) does not specify the rate of loading (displacement rate or load rate). In real structures, as the load increases and the tendons begin to fail, some load redistribution may be achieved between the structure elements. Therefore, the load carried by the damaged tendons may be reduced. This phenomenon may be considered comparable to the displacement control test.

Effect of Resin Type

Table 4 indicates that the E-tendons achieved 23% higher pseudoyielding load (P_y) on average compared with that of the V-tendons. Similarly, the elastic modulus (E) of the hybrid E-tendons was 3.7% higher than that of the V-tendons on average. The failure load (P_f) and failure strain (ε_f) of the hybrid E-tendons were only slightly higher than those of the V-tendons. However, the load loss at carbon failure of the E-tendons increased by 39% on average compared with that of the V-tendons.

The authors observed the hybrid effect phenomena reported by Manders and Bader (1981), Peijs and DeKok (1993), and Marom et al. (1978). The hybrid effect is when the failure strain of the

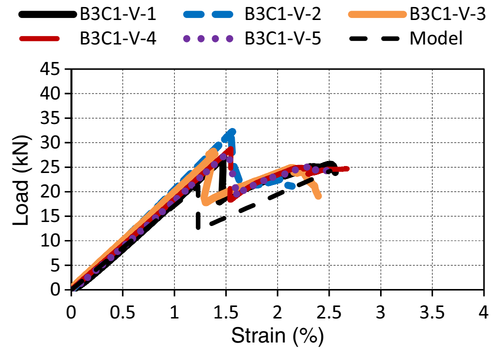


Fig. 7. Load-strain curve of B3C1-V

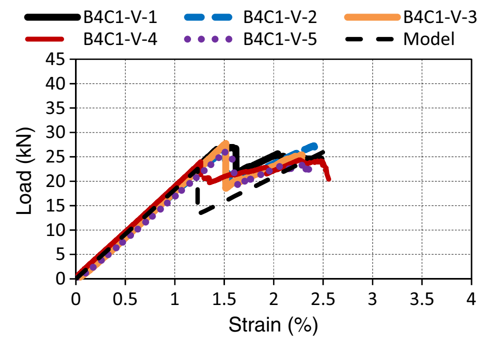


Fig. 8. Load-strain curve of B4C1-V

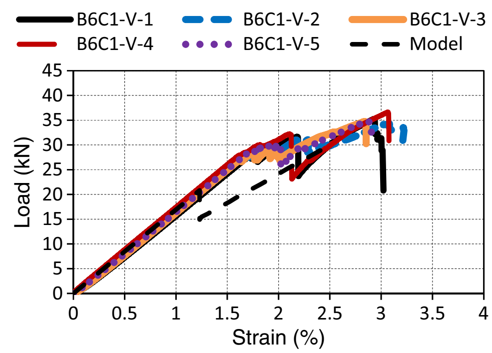


Fig. 9. Load-strain curve of B6C1-V

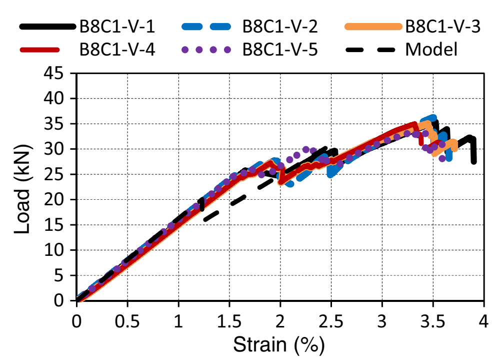


Fig. 10. Load-strain curve of B8C1-V

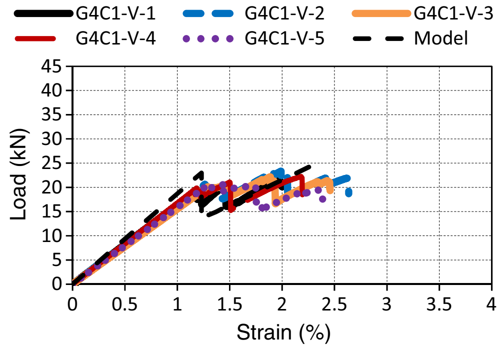


Fig. 11. Load-strain curve of G4C1-V

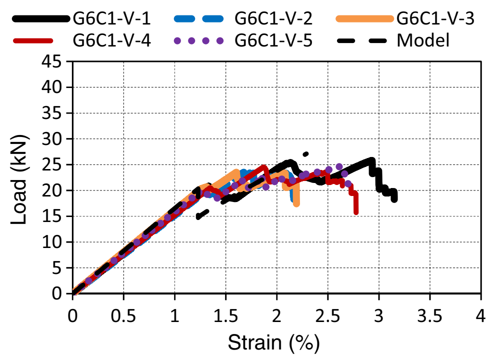


Fig. 12. Load-strain curve of G6C1-V

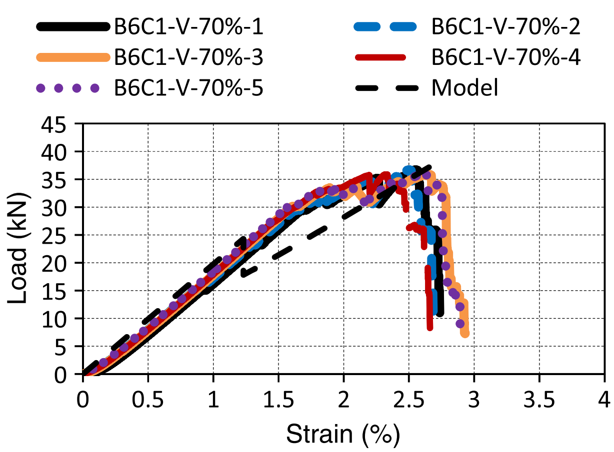


Fig. 13. Load-strain curve of B6C1-V-70%

lowest elongation fiber (carbon) appears to be greater in the hybrid composite compared with that in an all-carbon fiber composite. The average hybrid effect of E-tendons and V-tendons were 12 and 20%, respectively.

Effect of HE Fiber to LE Fiber Ratio

The results indicated that the load-strain behavior of B/C tendons could vary from brittle for hybrid tendons with $B/C = 1.0$ to a desired load-strain relationship similar to that of a steel rebar for hybrid tendons with $B/C = 8.2$. Similar to B/C tendons, changing the

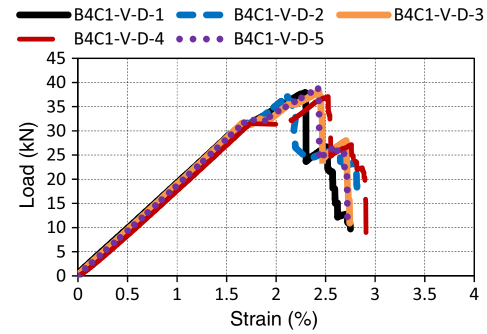


Fig. 14. Load-strain curve of B4C1-V-D

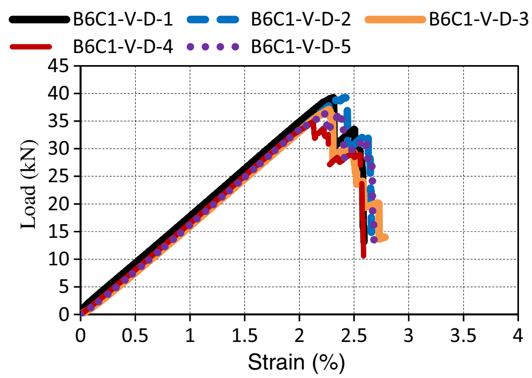


Fig. 15. Load-strain curve of B6C1-V-D

G/C ratio of G/C tendons from 2.5 to 3.7 or 6.8 could change brittle rupture to a ductile failure. Results revealed that increasing the B/C ratio effectively decreased the load drop at carbon failure. Table 4 shows that increasing the B/C ratio of the V-tendons from 1:1 to 2.7:1, 3.6:1, 6.5:1, and 8.2:1 decreases the load drop by 66, 79, 86, and 88%, respectively. Additionally, the failure strain, and hence the ductility, of B/C tendons is enhanced significantly by increasing the HE/LE fiber volume ratio. Moreover, the failure strain of V-tendons with B/C of 8.2:1 was 15% larger than the ultimate strain of the nonhybrid basalt tendons. Furthermore, the ratio of the failure load to pseudoyielding load increased nonlinearly with increasing B/C. The B8C1-V specimens achieved a failure load of 1.26 times its pseudoyielding load. Unfortunately, the tendon modulus decreased with increasing B/C ratio; however, its lowest modulus was still 15% higher than that of the nonhybrid basalt tendon modulus. The influence of the G/C ratio on load loss, failure strain, failure load to pseudoyielding load, and elastic modulus of the G/C hybrid tendons was similar to that of the B/C ratio on the B/C hybrid tendons.

The experimental results demonstrated that although the B3C1-V and G3C1-V tendons have the same carbon ratio, the failure of the former was more ductile than that of the G3C1-V tendons. Not only did the B6C1-V specimens exhibit a failure strain of 3.01%, which is 18% higher than that of the G6C1-V specimens, they also showed a higher P_f/P_y ratio. Moreover, the elastic modulus of the B6C1-V specimens was 9.1% higher than that of the G6C1-V specimens.

Overall, this experiment resulted in a new ductile hybrid B/C FRP tendon, B8C1-V, with a failure strain of 3.61%, which was 105% higher than its pseudoyielding strain, and with a failure load that was 1.35 times its pseudoyielding load. Such favorable characteristics were not achieved by the aramid/carbon hybrid tendons tested by Tamuzs et al. (1996). Although they utilized a braiding technique in the manufacture of their tendons, they reported that in the case of a polymerized braid (aramid) with a polymerized core (carbon), the loading curve did not differ from that of a specimen with unidirectional fibers only (Tamuzs 1996). Similar to Tamuzs et al. (1996), Young et al. (2007) attributed the brittle failure of their glass/carbon tendons to the sizable carbon ratio in the composite. However, Bakis et al. (2001) stated that glass/carbon hybrid tendons, having between 6 and 13% by volume carbon fibers dispersed in the cross section displayed a pseudoductile performance.

Effect of Fiber Volume Ratio

The authors investigated three fiber volume ratios (B6C1-V, B6C1-V-50%, and B6C1-V-70% specimens Table 1). As the results show, the load-strain behavior of the B/C hybrid tendon was significantly affected by the fiber volume ratio. The experiments indicated that the B/C hybrid tendon with 48% fiber volume, B6C1-V-50%, exhibited brittle failure, and a bilinear load-strain curve resulted in the case of the 59% fiber volume, as shown in Fig. 9. The B6C1-V-70% tendon achieved a load-stain curve similar to that of a high strength steel rebar, as shown in Fig. 13. It is well-known that when one filament fails its load transfers to the neighboring filaments, causing what is called a stress concentration. When the fiber volume fraction is high, the stress from the failed filament is distributed over more surrounding filaments; therefore, the stress concentration is relatively lower, resulting in a more progressive failure. Therefore, the load transfers from one filament to the next one smoothly, resulting in a more progressive failure.

The B6C1-V-70% tendons exhibited relatively higher modulus and smaller load loss in comparison with the B6C1-V tendons. Meanwhile, the failure strain of the B6C1-V tendons was 16% larger than that of the B6C1-V-70% tendons. Unfortunately, the B6C1-V-70% tendons have a smooth surface, which may decrease the bond with surrounding concrete when used in RC structures.

Effect of Fiber Arrangement

Figs. 14 and 15 show the tensile performance of the B/C hybrid tendons with carbon fiber dispersed over the cross section of the tendon. The load-strain response differed significantly from that of a similar specimen with carbon fiber placed at the center of the tendon. For B4C1-V-D, the load-strain relationship was linear up to a strain of approximately 1.65%, beyond which the strain increased slightly but without any substantial increase in load. The slope of the second stage of the response was small in comparison with the initial stiffness, and when the strain reached 2.35%, a sudden load loss of approximately 13.2 kN was soon followed by final rupture. In contrast, the B6C1-V-D load-strain curves were perfectly linear up to a strain of 2.27%, after which a noticeable load drop of about 8.4 kN occurred and the tendon failed at a strain of 2.52%. It is commonly believed that the failure strain of carbon filaments mainly depends on the position of the basalt fiber. Thus, it was expected that the failure strain of carbon filaments adjacent to basalt filaments would be higher than other carbon filaments. Fig. 6 shows that the number of carbon filaments near basalt fibers is higher in the case of dispersed carbon; therefore, the failure strain increased. Additionally, the failure of carbon filaments in the vicinity of basalt fibers caused rupture of some basalt filaments, which may have caused sudden failure of all of the remaining basalt filaments.

Validation of the Proposed Model

The modulus of elasticity and the pseudoyielding load of a hybrid composite can be computed accurately using the rule-of-mixtures, so they are not considered in this section. The authors took the relative displacement δ and shear strength τ_f from previous work (Wu et al 2002)as 0.3 mm and 0.008 GPa, respectively. They compared the accuracy of the proposed load-strain model of a hybrid FRP composite to predict the load drop at LE fiber rupture, failure load, and failure strain with the experimental data on hybrid FRP sheets/tendons carried out in this paper and those by Wu (2004), Kentaro et al. (2007), and Liang et al. (2004). This section summarizes the relevant details of the specimens tested by others. Wu (2004) and Kentaro et al. (2007) investigated hybrid FRP sheets that included higher modulus C7 ($E = 554$ GPa and $\varepsilon = 0.45\%$) carbon fiber sheets; higher strength C1 ($E = 243$ GPa and $\varepsilon = 1.74\%$) carbon fiber sheets, high-strength PBO (identified as P in Table 5 with $E = 260$ GPa and $\varepsilon = 1.60\%$) fiber sheets; and high ductility E-glass (identified as DEG in Table 5 with $\underline{E} = 80$ GPa and $\varepsilon = 1.88\%$) fiber sheets with different proportions. The layer numbers in Table 5 indicate the hybrid ratios of the various types of fibers. The hybrid FRP tension sheet specimens had a gauge length of 130 mm and width of 12.5 mm. Liang et al. (2004) investigated two types of glass/carbon hybrid rods with a nominal diameter and gauge length of 9.5 mm and 63.5 mm, respectively. The volume percentages of glass and carbon in the hybrid tendons were 48% and 13%, respectively. Liang et al. (2004) considered both randomly dispersed and core-shell hybrids, but the authors of this paper did not consider the tendons with randomly dispersed fibers in this paper. The modulus and failure strain of the glass and carbon were 89.6 GPa and 5% and 230 GPa and 2.1%, respectively.

Table 5 shows that the proposed model predicts the load drop at LE fiber rupture, failure load, and failure strain of a hybrid FRP sheet or tendon with good accuracy. The average prediction of the P_d, P_f, and ε_f were 0.83, 1.06, and 1.00, respectively. Table 5 also enables a comparison between the accuracy of the proposed model with that of the rule-of-mixtures in predicting the load drop, failure load, and failure strain of hybrid FRP tendons and hybrid FRP sheets. For the load drop prediction, both methods provide similar levels of accuracy in the case of hybrid tendons, because total delamination between the hybrid components was achieved as a result of the high loads carried by the tendons at the LE fiber rupture. However, for the hybrid FRP sheets, on average the proposed model predictions were approximately 0.91, whereas the prediction of the rule-of-mixtures was only approximately 0.53 on average. This is attributed to the small loads carried by the hybrid sheets at the LE fiber rupture, causing only partial delamination between the LE and HE fibers. The LE fibers continued to contribute to the hybrid stiffness after their rupture, as assumed by the proposed model. For the failure load prediction, the average predictions obtained by the proposed model in the case of tendons and sheets were 0.94 and 1.2, respectively, whereas the average predictions obtained by the rule-of-mixtures in the case of tendons and sheets were 0.79 and 1.06, respectively. The accuracy of the proposed model in predicting the failure load of hybrid sheets is slightly lower than that of the rule-of-mixtures method; this can be explained by the partial damage of the HE fibers at failure of the LE fibers in the case of hybrid sheets, which was approximately 10% as assumed by the proposed model. It should be less than this for the case of hybrid sheets. Finally, for the failure strain prediction, the proposed model exhibited excellent accuracy in comparison with the rule-of-mixtures method for both the hybrid tendons and hybrid sheets. The COV for the proposed model were relatively lower than those of the rule-of-mixtures method. Generally, the

Table 5. Comparison of Mechanical Properties with Prediction Models

		P_d (kN)					P_f (kN)					ε_f (%)				
Reference	Specimen ID	Exp	Model	Rule	Exp/ model	Exp/ rule	Exp	Model	Rule	Exp/ model	Exp/ rule	Exp	Model	Rule	Exp/ model	Exp/ rule
This paper (tendons)	B1C1-E	34	30	30.2	1.13	1.13	12	18.7	21.3	0.64	0.56	2.61	2.8	3.2	0.93	0.82
	B1C1-V	29	22.3	22.5	1.3	1.3	10	16.6	20.6	0.60	0.49	1.96	2.53	3.14	0.77	0.62
	B3C1-E	15	16.6	16.8	0.9	0.9	26	27.4	30.8	0.95	0.84	2.66	2.78	3.2	0.96	0.83
	B3C1-V	10	12.1	12.5	0.83	0.8	25	24.2	29.7	1.03	0.84	2.33	2.51	3.14	0.93	o.74
	B4C1-E	9	13.3	13.4	0.68	0.67	31	29.4	33.1	1.05	0.94	2.74	2.78	3.2	0.99	0.86
	B4C1-V	6	9.6	10.0	0.63	0.6	25	26.1	32.0	0.96	0.78	2.42	2.51	3.14	0.96	0.77
	B6C1-V	4.2	5.9	6.35	0.71	0.66	35	30.1	36.9	1.16	0.95	3.01	2.51	3.14	1.20	0.96
	B8C1-V	3.6	4.7	5.2	0.77	0.69	35	31.0	38.0	1.13	0.92	3.61	2.51	3.14	1.44	1.15
	B6C1-V-70%	2.9	6.5	7.4	0.45	0.39	36	38.8	43.1	0.93	0.84	2.59	2.77	3.14	0.94	0.82
	G4C1-V	5.1	9.8	10.1	0.52	0.50	22	23.0	25.6	0.96	0.86	2.34	2.1	2.4	1.11	0.98
	G6C1-V	3.1	5.7	6.1	0.54	0.51	24	25.5	28.4	0.94	0.85	2.55	2.1	2.4	1.21	1.06
Wu (2004) (sheets)	C1C7(1:1)	2.57	2.58	4.45	0.996	0.58	5.73	5.23	5.90	1.10	0.97	—	1.37	1.74	—	—
	PC7(1:1)	2.21	2.46	4.50	0.90	0.49	5.44	5.97	6.70	0.91	0.81	—	1.26	1.60	—	—
	C1C7(1.5:1)	2.17	2.4	4.50	0.90	0.48	8.08	7.33	8.25	1.10	0.98	1.37	1.4	1.74	0.98	0.79
	C1C7(2:1)	0.9	1.47	4.50	0.61	0.2	11.9	10.5	11.7	1.13	1.01	1.35	1.36	1.74	0.99	0.78
	PC7(2:1)	0.75	1.24	4.46	0.60	0.17	10.0	11.9	13.3	0.84	0.75	1.28	1.22	1.60	1.05	0.8
Kentaro et al. (2007) (sheets)	C1C7(0.4:1)	3.11	2.7	3.40	1.15	0.91	2.61	1.68	1.89	1.55	1.38	0.92	1.08	1.48	0.85	0.62
	C1C7(1.2:1)	1.87	1.88	3.40	0.99	0.55	7.9	5.1	5.70	1.55	1.39	1.25	1.11	1.48	1.13	0.84
	PC7(0.4:1)	3.27	2.68	3.38	1.22	0.96	3.23	2	2.24	1.62	1.44	1.01	1.1	1.46	0.92	0.69
	PC7(1.2:1)	1.16	1.81	3.38	0.64	0.34	7.8	6.01	6.73	1.30	1.16	1.2	1.11	1.46	1.08	0.82
	DEGC7(1:0.5)	0.8	1.06	1.69	0.75	0.47	1.84	1.97	2.22	0.93	0.82	0.5	1.46	1.88	0.34	0.27
	DEGC7(2:0.5)	1.14	0.9	1.68	1.27	0.68	4.44	3.91	4.44	1.14	1.0	1.69	1.51	1.88	1.12	0.9
Liang et al. (2004) (tendons)	C13G48	23.5	44.5	44.5	0.53	0.53	84.4	96.7	153	0.87	0.55	3.5	3.17	5	1.1	0.7
Average accuracy	—	—	—	—	0.83	0.63	—	—	—	1.06	0.92	—	—	—	1.00	0.80
Coefficient of variance (%)	—	—	—	—	31	43	—	—	—	24	27	—	—	—	21	23

Note: Pd = load drop at carbon rupture; Pf = failure load; ε_f = failure strain; Exp = experimental value.

proposed model more accurately predicted the mechanical properties of hybrid tendons/sheets in comparison with the rule-of-mixtures method.

Conclusions

This paper presented an iterative analytical model for predicting the tensile performance of hybrid FRP composites, taking into account the contribution of the LE fiber to the hybrid stiffness after its failure. The proposed model accurately identified the load drop at the LE fiber rupture, the failure load, and the failure strain of hybrid FRP composites. Moreover, the authors investigated the mechanical properties of all basalt, all carbon, and hybrid B/C and G/C tendons. The use of epoxy resin in fabricating FRP composite tendons indicated superior performance in comparison with vinylester resin. Experiments identified that the load-strain behavior of B/C hybrid tendons was significantly affected by the fiber volume ratio. This paper presented a new ductile B/C FRP hybrid tendon, with a failure strain of 3.61%, which was approximately 105% higher than its pseudoyielding strain, and a failure load that was 1.35 times its pseudoyielding load.

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Notation

The following symbols are used in this paper:

A_1 = cross-sectional area of HE fiber;
A_2 = cross-sectional area of LE fiber;
C1 = high-strength carbon;
C7 = high-modulus carbon;
E_1 = modulus of elasticity of HE fiber;
E_2 = modulus of elasticity of LE fiber;
f_1 = tensile strength of LE fibers;
G_f = interfacial fracture energy;
L_b = specimen length where LE and HE fibers are bonded;
L_t = total length of specimen;
L_{ub} = specimen length where LE and HE fibers are unbonded;
P_d = load drop at the rupture of the lowest elongation fiber;
P_f = failure load;
P_y = pseudoyielding load;
S = width of the interfacial surface;
V_f = fiber volume ratio;
v_1 = volume ratio of HE fiber;
v_2 = volume ratio of LE fiber;
β = ratio of LE fiber to HE fiber stiffness;
δ = relative displacement between adherents;
δ_f = maximum relative displacement between adherents;
ε_1 = failure strain of HE fiber;
ε_f = failure strain;
ε_y = pseudoyielding strain;
σ_1 = normal stress at HE fiber;
τ = shear stress of adhesive layer;
τ_f = shear strength of adhesive layer;

ϕ = reduction factor of HE fiber strength;
ψ = ratio between bonded length to total length of specimen;
ψ_{min} = minimum bonded length ratio; and
ω = percentage of load transferred to HE fiber at end of specimen.

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